VIKRANT GUPTA

Observational Astronomy | Galaxy Evolution | Stellar Populations

B.S.–M.S. Physics, IISER Pune, India • vikrant.rajkumargupta@students.iiserpune.ac.in

RESEARCH PROFILE

Undergraduate (BS-MS, IISER Pune) with hands-on experience in observational extragalactic work: multiwavelength catalog analysis of thousands of galaxies and a study of bar structure in interacting systems at IIA. Core interests are galaxy evolution and radio astronomy, with a growing focus on pulsars. Seeking a long-term research collaboration, from a summer project to a master's thesis. Seeking a long-term research collaboration, starting with a summer 2027 internship.

RESEARCH EXPERIENCE

Summer Research Fellow, IIA Bengaluru May–Jul 2026

Galaxy Bars in Interacting Systems | Prof. Mousumi Das, Ms. Saili Keshri, Dr. Sudhanshu Barway

- Screened 6,292-object Arp–Madore catalogue by visual morphology on DECaLS DR9/DR10, counting a system as interacting if it showed tidal features or a companion within $6'$ with matching redshifts (SIMBAD, NED), then narrowed 372 candidates to 75 systems with bars confirmed by isophote fitting.
- Switched from optical r-band to VISTA VIRCam K_s-band ($\sim$2.15 µm) imaging after dust extinction and tidal star formation stopped the isophote fits from converging, and to trace the older stellar mass that makes up the bar.
- Built automated Python/Photutils workflows with star and companion masking, local sky subtraction, and elliptical-isophote fitting, identifying bars from the joint behaviour of the ellipticity and position-angle profiles and measuring disk orientation at the $\mu_{K_s} = 20\,\mathrm{mag\,arcsec^{-2}}$ (R_{20}) isophote.
- Deprojected observed bar lengths to intrinsic face-on lengths for $20^\circ \leq i \leq 60^\circ$ and normalised by R_{20} (median $L_{\mathrm{bar}}/R_{20} \approx 0.47$).
- Computed integrated K_s-band luminosities, stellar masses (constant $M/L = 0.6$), and pair mass ratios q for 62 measurements across 57 systems, using Vega-to-AB conversion and redshift-based distances, with calibration uncertainties propagated.
- Found dual-barred pairs only among major mergers ($q \geq 0.3$). Normalised bar length showed a weak positive trend with $\log q$ for single-barred systems ($r \approx 0.28$) and a statistically insignificant negative trend for dual-barred pairs ($\rho \approx -0.07$, $N = 10$).

Independent Research Project Dec 2025

Integrated Galaxy SED Flux vs. Stellar Population Age | Prospector SPS Framework

- Designed a controlled numerical experiment in Prospector to isolate how the wavelength-integrated (UV to mid-IR), rest-frame flux of a single stellar population fades with age. Held mass ($10^{10}\,M_\odot$), metallicity, and dust fixed, sampled 6,000 log-spaced ages from 0.01 to 10 Gyr, and normalised the flux to the 1 Gyr model of each configuration.
- Fitted an approximate power law $R(t) \propto t^{-\alpha}$ by least squares in log–log space ($\alpha \approx 1.07$ at baseline) and repeated it across $-1.7 \leq \log(Z/Z_\odot) \leq +0.18$ and $0.3 \leq$ `dust2` ≤ 2.5. The exponent stayed within $0.94 \lesssim \alpha \lesssim 1.08$, steepening weakly at low metallicity ($\Delta\alpha \lesssim 0.02$) and flattening with dust ($1.057 \rightarrow 0.940$), so it is stable within this model grid.
- Scope: single-age population and integrated flux only, with no star-formation histories, observational data, or parameter inference, so the relation does not constrain real galaxies. Excluded the dust-free case, whose non-smooth ratios are attributable to numerical effects, and engaged critically with Bayesian inference and parametric vs. non-parametric star-formation histories in Prospector.

LEADERSHIP & OUTREACH

Co-founder & Head, Aerodynamics & Mechanical Dept. IISER Pune

Kshitij, IISER Pune Rocketry Team

- Co-founded the student rocketry team and lead its aerodynamics and mechanical department. A miniaturised vehicle reached its 100 m target apogee, and the first full-scale launch is targeted for January 2027.

Speaker, Science Club IISER Pune

Talk on Research in Astronomy

- Gave a talk sharing what hands-on astronomy research has taught me, to guide and motivate juniors toward research.

Volunteer Teacher, Disha Initiative Education Outreach

Teaching Underprivileged Children

- Taught physics concepts such as gravity in simple, intuitive terms to around 35 underprivileged children.

EDUCATION

IISER Pune 2024–Present

BS-MS Dual Degree, Physics

3rd Year | CGPA: 9.2/10.0

Coursework: Quantum Mechanics I, Statistical Mechanics I, Electrodynamics, Mathematical Methods, Probability & Statistics, Computing & Data Analysis, Group Theory, Real Analysis I, Discrete Structures, Classical Mechanics, Thermal & Statistical Physics.

Upcoming: Astronomy courses (next semester).

RESEARCH INTERESTS

- Galaxy evolution
- Observational surveys and photometry
- Stellar populations

KEEN TO EXPLORE

- Radio astronomy
- Early universe
- Simulation-based approaches

TECHNICAL SKILLS

- **Python:** NumPy, SciPy, Pandas, Matplotlib, Astropy, Photutils, FITS handling, Bash scripting
- **Astrophysics:** Prospector (SPS/SED, Bayesian inference), deprojection, isophotal photometry, ellipse fitting, stellar masses
- **Data:** VISTA VIRCam K_s, DECaLS DR9/DR10, SIMBAD, NED
- **Tools:** LaTeX
- **Engineering:** Autodesk Fusion 360, CAD, CFD; FEA (buckling, shock, static stress, and frequency analysis)

HONOURS & AWARDS

INSPIRE Scholarship (SHE)
DST, Govt. of India (2024–Present)
Awarded for ranking in the top 1% nationally.